Electricity supplies

ELECTRICITY IS SENT FROM POWER STATIONS FOR USE IN HOMES AND WORKPLACES VIA A HUGE NETWORK OF CABLES.

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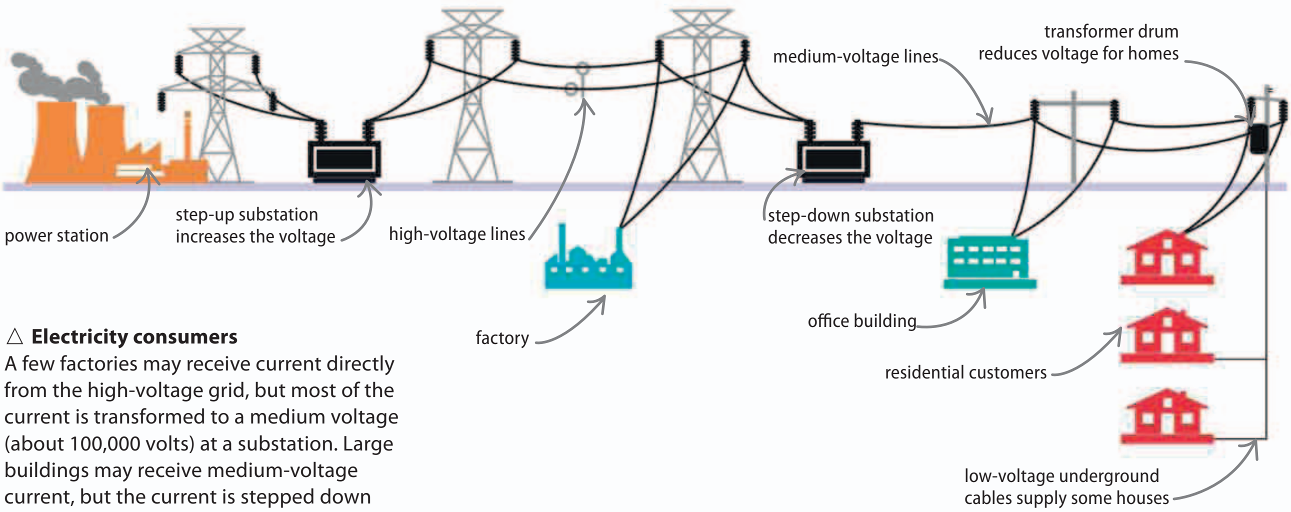
Almost all the electricity used in homes, offices, and factories is generated at large power stations far from where people live. It is sent across country in a power grid, before being transformed into a usable current suitable for domestic use.

Power grid

Electricity is generated as an alternating current (AC). This is boosted to several hundred thousand volts by a transformer before it enters the power grid. The high voltage reduces the amount of energy lost as heat as currents travel along wires hundreds of kilometers long. Burying high-voltage lines is very expensive, so most of the power grid is made up of lightweight aluminum cables suspended from pylons, high in the air for safety.



△ **Electricity consumers**
A few factories may receive current directly from the high-voltage grid, but most of the current is transformed to a medium voltage (about 100,000 volts) at a substation. Large buildings may receive medium-voltage current, but the current is stepped down again by transformers to between 110 and 250 volts before reaching homes.

Future power grids may use superconductors to carry **ten times as much current** as today's cables.

REAL WORLD

Power cuts

When the power grid fails, it results in a power cut. No current arrives at homes and offices, and the lights—and everything else—go off. A power cut can be caused by a simple failure of a transformer or a cable being damaged by a storm. However, huge power cuts have also been caused by solar storms, where a surge of charged particles from the Sun overloads the grid, causing it to shut down.